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A TCN-BiLSTM-based approach for remaining useful life prediction of PEMFC☆

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ABSTRACT

Against the global energy crisis and environmental pollution, proton exchange membrane fuel cell (PEMFC) is key green energy device, yet their large-scale application is impeded by inaccurate remaining useful life (RUL) prediction and standalone deep learning model flaws. To resolve these issues, this study proposes a parallel TCN-BiLSTM framework for high-precision PEMFC RUL forecasting. A composite denoising model (VMD-KF-MF) purifies voltage signals while retaining degradation features; residual connections boost long-sequence capture of temporal convolutional network (TCN), and an improved artificial lemming algorithm (IALA) optimizes bidirectional long short-term memory (BiLSTM) weights; point-wise mean squared error (MSE) integrates sub-model outputs. Compared with existing models, the model achieves a prediction accuracy of 99.35%, and its stability and robustness are greatly improved, advancing PEMFC prognostic technology, enabling reliable predictive maintenance, cutting operational costs, and accelerating PEMFC industrialization.

1. Introduction

With the worsening global energy crisis and environmental pollution, exploring green and sustainable energy sources has become imperative to meet global energy conservation and emission reduction goals. PEMFC, a zero-emission, high-efficiency device that directly converts chemical energy into electrical energy, exhibits great application potential in electric vehicles, distributed power, and mobile power sources, serving as a key future alternative to traditional fossil fuels [1,2]. Recent advances in PEMFC materials and devices—including high-efficiency single-atom Pt catalysts, self-healing proton exchange membranes, and fuel cell-supercapacitor hybrids—have expanded its application scenarios, yet their complex degradation mechanisms pose new challenges for RUL prediction [3,4].

However, PEMFCs' short service life leads to high costs, hindering their widespread deployment [5]. This is caused by internal component aging (membrane degradation, catalyst deactivation, gas diffusion layer performance loss) and external factors like temperature and humidity, which collectively drive gradual performance degradation to failure [6,7]. The complexity of these factors, coupled with the fuel cell's internal chemical energy instability, makes RUL prediction challenging. Developing effective strategies for accurate RUL prediction is therefore critical to provide a theoretical basis for assessing subsequent usability.

Deep learning, renowned for its strong data processing capabilities and ability to capture complex nonlinear relationships [8,9], has become a key paradigm for PEMFC RUL prediction. Unlike traditional methods, it eliminates the need for in-depth knowledge of PEMFCs' internal chemical mechanisms [10], instead autonomously learning latent nonlinear correlations from large datasets to discern the mapping between performance degradation and influential features, enabling accurate RUL prognostication [11]. However, if the RUL prediction error is high, it will lead to phenomena of over-maintenance or maintenance insufficiency. For instance, a 10% Health Indicator (HI) loss can increase the operation and maintenance costs of commercial vehicle fuel cell systems by 15%–20%. If the RUL prediction error exceeds 15%, it will result in misjudgment of the aging state of the core components of the PEMFC, which significantly raises the economic costs and safety risks of the PEMFC system [12]. For high-value novel PEMFC components, such prediction errors may even cause irreversible damage and higher economic losses.

[13] used XGBoost for feature selection and combines self-attention with LSTM for PEMFC performance degradation prediction. While self-attention enhances real-data prediction accuracy, LSTM risks overfitting
—requiring strict data screening and poorly handling extreme operating conditions. [14] applied a TCN-equivalent circuit hybrid model for

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Table 1
Range of physical parameters for data collection environment.

Parameter	Controlled scope
Cooling temperature	20 °C – 80 °C
Cooling flow rate	0–10 l/min
Gas temperature	20 °C – 80 °C
Gas humidity	0–100%RH
Air flow rate	0–100 l/min
Hydrogen flow rate	0–30 l/min
Gas pressure	0–2 bars
Battery current	0–300 A

RUL prediction. TCN captures long-sequence dependencies via dilated convolutions but lacks short-sequence temporal modeling flexibility (prone to local information loss) and adapts poorly to bidirectional contextual tasks due to unidirectional causal convolutions. [15] proposed a BiLSTM-Self-attention-DNN hybrid model, which integrates voltage decay sequences and dynamic operating features for multiscale degradation capture. However, BiLSTM's recurrent structure suffers from information attenuation with longer sequences, hindering long-distance dependency capture and reducing performance.

Based on the above analysis, this paper proposes a TCN-BiLSTM model for PEMFC RUL prediction. First, variational mode decomposition (VMD), Kalman filter (KF), and median filter (MF) are combined for stepwise shallow-to-deep denoising of complex signals. Second, the IALA algorithm optimizes BiLSTM weights, with TCN and BiLSTM jointly performing sequence prediction; final results are screened via error metrics. Compared to mainstream particle swarm optimization (PSO) and grey wolf optimizer (GWO) algorithms (prone to unstable iterations when optimizing BiLSTM), the proposed model achieves significant performance improvements. Its main innovations are summarized as follows:

- Exponential decay for ALA balances exploration–exploitation, curbs premature convergence and improves search performance.
- IALA-optimized BiLSTM weights accelerate parameter identification and enhance prediction accuracy.
- The parallel TCN-BiLSTM predictor fully integrates their advantages and boosts prediction performance.

Experimental results show the proposed model achieves 99.35% prediction accuracy, outperforming standalone TCN and BiLSTM significantly.

2. PEMFC system

PEMFC is an electrochemical reaction, which enables selective conduction through a proton exchange membrane [16]. Its chemical reaction formula is as follows:

$$\begin{cases} H_2 \rightarrow 2H^+ + 2e^- \\ \frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O \\ H_2 + \frac{1}{2}O_2 \rightarrow H_2O \end{cases} \tag{1}$$

This study uses the public dataset from FCLAB research consortium, acquired under strictly controlled conditions to investigate how PEMFC operating conditions affect performance degradation. The tested PEMFCs include 5 stacks (each 100 cm^2, nominal current density 0.70 A/cm^2), with the acquisition environment's physical parameter ranges shown in Table 1.

Remark 1. To address the original dataset's large size, 500 initial voltage signals were selected as experimental data—preserving the mapping between original characteristic signals and time series. This choice is justified by two factors: voltage signals are the most intuitive for PEMFC RUL observation, and it avoids overfitting risks in model training induced by excessive noise from large datasets.

3. Basic theories

3.1. Variational mode decomposition

VMD, an adaptive signal processing method, decomposes complex multi-component chaotic signals into intrinsic mode functions (IMFs) with specific frequency bandwidths [17]. The corresponding formula is as follows:

$$\begin{cases} \min \left\{ \sum_{k=1}^{K} \left\| \partial t \left[\left(\delta(t) + u_k(t) \right) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ s.t. \sum_{k=1}^{K} u_k(t) = f(t) \end{cases} \tag{2}$$

where ∂t is partial derivative; $\delta(t)$ is Dirac distribution function; K is total number of decomposed components; $u_k(t)$ is IMF signal; ω_k is center frequency; $\|\cdot\|_2^2$ is square of the L_2 norm; $f(t)$ is original signal.

Introducing the Lagrange multiplier λ and quadratic penalty term a transforms (2) into an unconstrained extremum problem, with the mathematical expression as follows:

$$\begin{aligned} L\left(\{u_k\}, \{\omega_k\}, \lambda\right) = & a \sum_{k=1}^{K} \left\| \partial t \left[\left(\delta(t) + \frac{j}{\pi t} \right) u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ & + \left\| f(t) - \sum_{k=1}^{K} u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{k=1}^{K} u_k(t) \right\rangle \end{aligned} \tag{3}$$

where $\langle \cdot \rangle$ is the inner product operation.

Update formulas for each IMF, its center frequency, and the Lagrange multiplier are as follows:

$$\begin{cases} \hat{u}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{u}_k^{n+1}(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha\left(\omega - \omega_k^n\right)^2} \\ \omega_k^{n+1} = \frac{\int_0^\infty \omega \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega}{\int_0^\infty \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega} \\ \hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \gamma \left[\hat{f}(\omega) - \sum_{k=1}^{K} \hat{u}_k^{n+1}(\omega) \right] \end{cases} \tag{4}$$

where $\hat{f}(\omega), \hat{u}_k(\omega), \hat{\lambda}(\omega)$ are Fourier transforms of the original signal, IMF signal, and Lagrange multiplier, respectively; γ is Lagrange multiplier update step size.

Iteration terminates when the stopping condition is satisfied, with the formula as follows:

$$\sum_{k=1}^{K} \left(\frac{\left\| \hat{u}_k^{n+1} - \hat{u}_k^n \right\|_2^2}{\left\| \hat{u}_k^n \right\|_2^2} \right) < \varepsilon \tag{5}$$

where ε is a number approaching zero.

This paper uses mutual information and Pearson correlation coefficient to validate IMFs decomposed from the original signal [18,19]. Selected for their ability to quantify correlations between continuous/discrete variables and capture linear/nonlinear relationships, the formulas are as follows:

$$\begin{cases} h = \sum_{x \in X} \sum_{y \in Y} P(x,y) \log_2 \left(\frac{P(x,y)}{P(x) \cdot P(y)} \right) \\ r = \frac{\sum_{x \in X} \sum_{y \in Y} (x - \bar{X})(y - \bar{Y})}{\sqrt{\sum_{x \in X} (x - \bar{X})^2} \sqrt{\sum_{y \in Y} (y - \bar{Y})^2}} \end{cases} \tag{6}$$

where X, Y are the two IMF components; x, y are data points in X, Y; $\bar{X}, \bar{Y}$ are their means; $P(x), P(y)$ are their interval frequencies; $P(x,y)$ is co-occurrence frequency of their discrete intervals; h, r are mutual information value and Pearson correlation coefficient, respectively.

To normalize the mutual information coefficient to the range [0, 1], the formula is as follows:

$$NM(h) = \frac{h}{\min\left(-\sum_{x \in X} P(x) \log_2 P(x), -\sum_{y \in Y} P(y) \log_2 P(y) \right)} \tag{7}$$

3.2. Kalman filter

KF estimates the true system state from noisy measurement data by iteratively correcting measurements for higher accuracy [20].

3.2.1. One-dimensional system model

For one-dimensional feature data (voltage signals in this study), KF converts it into a linear time-invariant system with state and observation equations.

(1) Equation of state

It characterizes the signal's true value law, with the formula as follows:

$$x_k = x_{k-1} + \upsilon_{k-1} \tag{8}$$

where x_k is true value at time k; υ_{k-1} is process noise at time $k-1$ ($\upsilon \sim N(0, Q)$, Q is process noise variance, reflecting the true signal's stationarity).

(2) Equation of observation

It describes the noisy observation process, where the measured value is the superposition of the true value and noise. The formula is as follows:

$$z_k = x_k + \varsigma_k \tag{9}$$

where z_k is measured value (original noisy data) at time k; ς_k is measurement noise at time k ($\varsigma \sim N(0, R)$), R is measurement noise variance (reflecting original data noise intensity).

3.2.2. Iterative filtering process

KF achieves denoising via "prediction-update" iterations; specific steps are as follows:

(1) Prediction stage

Predicting the current signal state from the previous optimal estimate provides a denoising reference.

(1.1) State prior estimation

Assuming stable signal variation, the previous optimal estimate serves as the current predicted value, with the formula as follows:

$$\hat{x}_k^- = \hat{x}_{k-1}^- \tag{10}$$

where $\hat{x}_k^-$ is prior estimate (predicted value) at time k; $\hat{x}_{k-1}^-$ is optimal estimate (denoising result) at time $k-1$.

(1.2) Prior error covariance

Due to process noise, the predicted result has greater uncertainty than the previous moment. The formula is as follows:

$$P_k^- = P_{k-1} + Q \tag{11}$$

where P_k^- is predicted value error covariance; P_{k-1} is posterior error covariance.

(2) Update phase

Correcting the prediction using the current measurement yields the denoised optimal estimate.

(2.1) Kalman gain

It dynamically balances the weights of the true value and prediction. The formula is as follows:

$$K_k = \frac{P_k^-}{P_k^- + R} \tag{12}$$

where $K_k \in (0, 1)$ is the Kalman gain.

(2.2) Posterior state estimation

Fusing the predicted and measured values outputs the denoised signal. The formula is as follows:

$$\hat{x}_k = \hat{x}_k^- + K_k \cdot (z_k - \hat{x_k^-}) \tag{13}$$

where $(z_k - \hat{x_k^-})$ is the measurement residual.

(2.3) Posterior error covariance

Update the optimal estimate's uncertainty for the next moment's prediction. The formula is as follows:

$$P_k = (1 - K_k) \cdot P_k^- \tag{14}$$

VMD-decomposed IMFs are individually denoised via KF. The advantage is their mutual independence: denoising avoids eliminating the overall signal's true data due to incomplete decomposition, enabling more thorough denoising.

Signal-to-noise ratio (SNR) is used as the evaluation index; the formula is as follows:

$$\sigma_{SNR} = 10 \log_{10} \left(\frac{\sum_{k=1}^{N} z_k^2}{\sum_{k=1}^{N} (z_k - \hat{x}_k)^2} \right) \tag{15}$$

Denoising results of each IMF and their reconstructed signals are shown in Figs. 1 and 2.

3.3. Median filter

MF is a nonlinear order-statistics-based filtering method [21]. Its core principle is to replace the central value of a signal's local window with the window's median to suppress noise. For 1D signal $x(n)(n = 0, 1, 2, \ldots, N)$ with noisy version $y(n) = x(n) + \eta(n)$ ($\eta(n)$ is noise), MF's processing steps are as follows:

(1) Sliding window definition

Select a window of length L centered at current processing point $y(i)$; the covered signal sequence is $\{y(i-k), y(i-k+1), \ldots, y(i), \ldots, y(i+k-1), y(i+k)\}$, where i is window center index and k is window half-width.

(2) Sorting and median calculation

Sort the L window signal values in ascending order to form the ordered sequence $\{y_{(1)}, y_{(2)}, \ldots, y_{(L)}\}$ ($y_{(1)} \le y_{(2)} \le \cdots \le y_{(L)}$); the window median is the middle element $y_{(k+1)}$.

(3) Filter output

The window center's denoising result is the window median: $\hat{x}(i) = median\{y(i-k), \ldots, y(i+k)\} = y_{(k+1)}$.

Finally, KF-processed signals are further denoised via MF to yield purer denoised data, as shown in Fig. 3.

Remark 2. Experimental results demonstrate that a window size of 9 effectively suppresses most sharp noise components. Nevertheless, certain discrepancies and residual interference persist relative to the target signal; thus, secondary MF is implemented to eliminate subtle noise points and further refine signal quality.

To eliminate subtle noise rigorously, five windows of sizes 5, 10, 15, 20, and 25 are used for secondary processing; the Pearson correlation coefficient method serves as the evaluation index to identify the optimal processed signal.

The selected window sizes effectively eliminate sharp noise, and the Pearson correlation coefficient confirms processed signals align with the original signal; thus, a window size of 15 is chosen to reconstruct the final signal, as shown in Fig. 4.

3.4. Temporal convolutional network

TCN is a network architecture specifically designed for time-series data [22]. It overcomes the parallel computing limitations of traditional recurrent neural network (RNN); additionally, stacked convolutional layers enable it to effectively capture local feature dependencies in data.

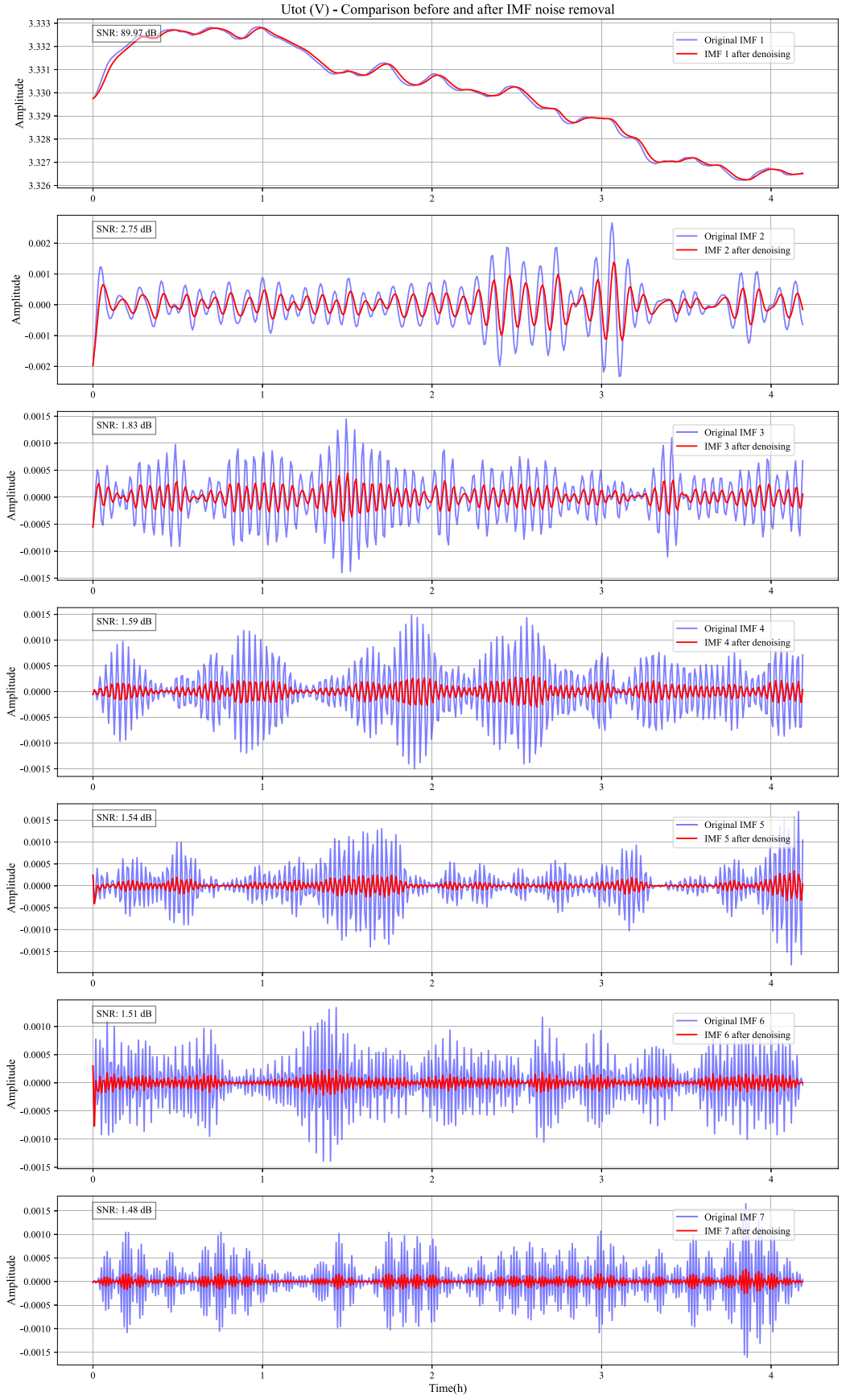


Fig. 1. Each IMF conducts KF denoising.

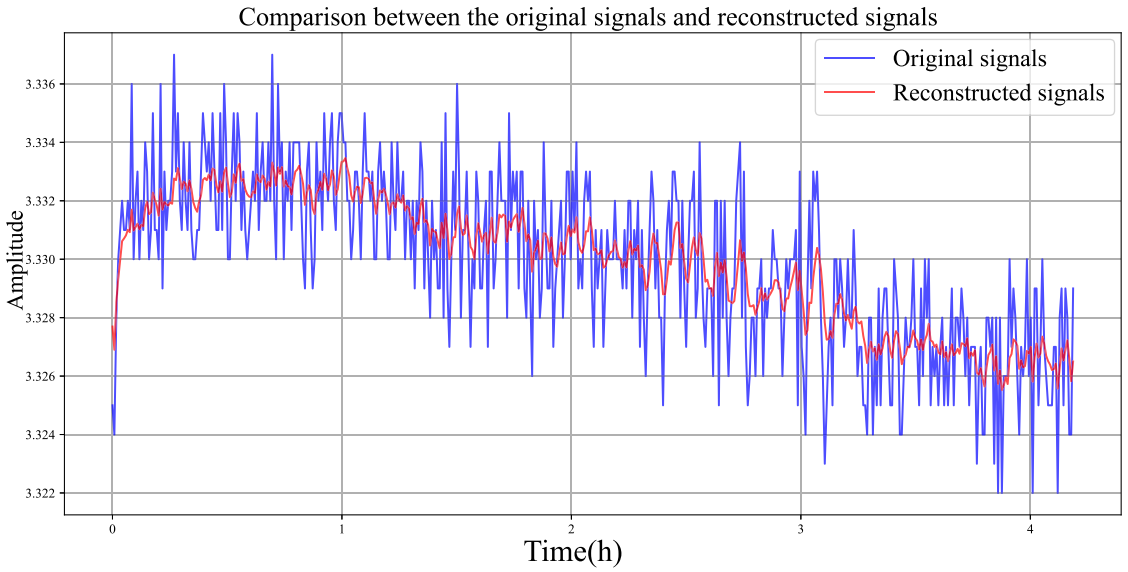


Fig. 2. KF reconstruction signal.

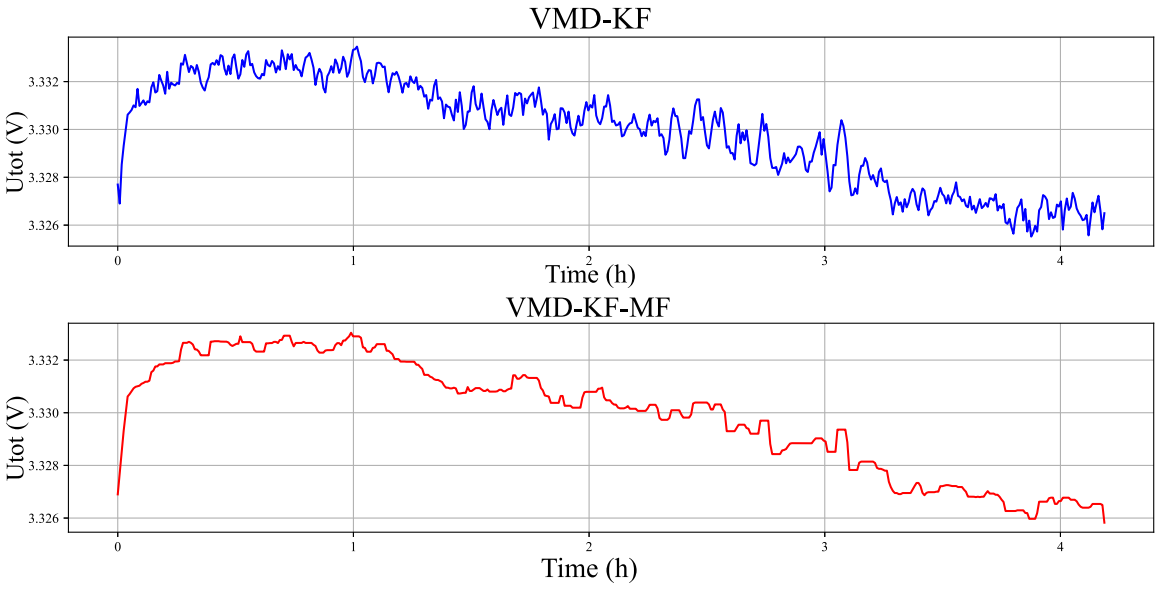


Fig. 3. Denoising image with VMD-KF-MF.

3.4.1. Causal convolution

TCN employs causal convolution to ensure the model only uses past inputs for time-series prediction tasks. For 1D time-series input $x \in \Omega^T$ (T is sequence length), the mathematical definition of causal convolution is as follows:

$$y_t = \sum_{k=0}^{K-1} \phi_k x_{t-k} \tag{16}$$

where y_t is convolution output at time t, K is convolution kernel size, ϕ_k is convolution kernel weight, x_{t-k} is input at time $t-k$ (set to 0 if $t-k<0$).

3.4.2. Dilated convolution

To capture long-term dependencies, TCN integrates dilated convolution to expand the convolution kernel's receptive field. Specifically, inserting "zero rows" between kernel element rows enlarges the receptive field without changing kernel size; after L dilated convolution layers, the total receptive field G is expressed as:

$$G = 1 + \sum_{l=0}^{L-1} \cdot d_l \tag{17}$$

where l is network layer count, d_l is dilation rate of the lth network layer.

Dilated convolution output is given as follows:

$$y_t = \sum_{k=0}^{K-1} \phi_k x_{t-d\cdot k} \tag{18}$$

where $d \geq 1$ is the dilation rate.

3.4.3. Residual connection

To mitigate TCN's gradient explosion and vanishing issues, residual connections are integrated into its architecture; additionally [23], these connections can accelerate model learning and boost prediction accuracy to some extent. The TCN architecture is shown in Fig. 5.

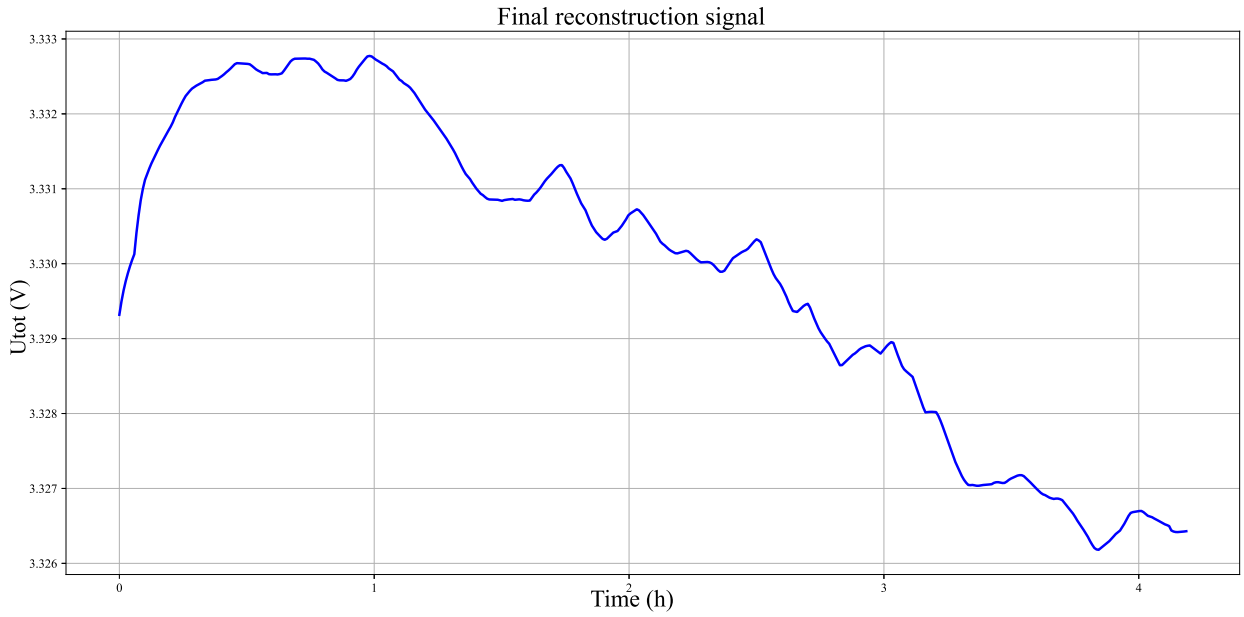


Fig. 4. Final reconstructed signal.

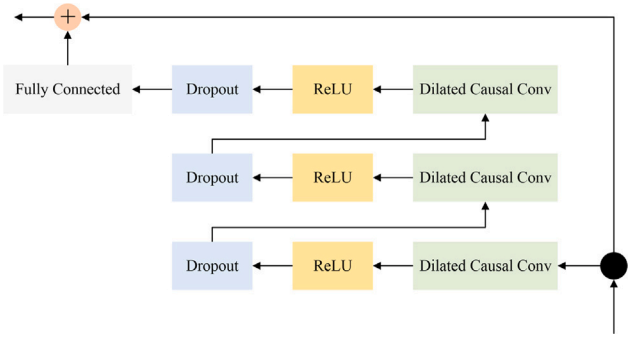


Fig. 5. Network structure of TCN.

3.5. Bidirectional long short-term memory network

BiLSTM is an improved recurrent neural network for sequence data processing [24].

(1) Forget gate

The forget gate determines which cell state information to discard. It takes the prior hidden state h_{t-1} and current input x_t, concatenates them, and applies the sigmoid activation function to produce a vector f_t. Each f_t element corresponds to a C_{t-1} element: values near 0 mean most corresponding cell state information is forgotten, while values near 1 mean most is retained. Its formula is as follows:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \tag{19}$$

where W_f is forget gate weight matrix, b_f is bias term, σ is sigmoid function.

(2) Input gate

The input gate updates the cell state by determining which new information to incorporate, and it comprises two components. The first component uses the sigmoid function to select update information, with its mathematical expression given as follows:

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \tag{20}$$

where W_i is input gate weight matrix, b_i is bias term.

The second component generates a new candidate value vector $\widetilde{c}_t$ via the tanh function: it takes h_{t-1} and x_t, passes them through a fully connected layer and the tanh function, with its formula given as follows:

$$\widetilde{c}_t = \tanh\left(W_c \cdot \left[h_{t-1}, x_t\right] + b_c\right) \tag{21}$$

where W_c is candidate weight matrix, b_c is bias term.

Updated cell state c_t is given by the following formula:

$$c_t = f_t * c_{t-1} + i_t * \widetilde{c}_t \tag{22}$$

(3) Output gate

The output gate controls the final output: it takes h_{t-1} and x_t, passes them through a fully connected layer and sigmoid function to generate a 0–1 value that selects which cell state parts to output. Its formula is as follows:

$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}, x_t\right] + b_o\right) \tag{23}$$

where W_o is output gate weight matrix, b_o is bias term.

Finally, the updated cell state c_t is passed through the tanh function and multiplied by the output gate's output to yield the final hidden state h_t. Its formula is as follows:

$$h_t = o_t * \tanh\left(c_t\right) \tag{24}$$

Via these gating operations, LSTM effectively retains key information and flexibly updates/outputs information when processing sequence data. For BiLSTM, output is jointly determined by forward LSTM output $\overrightarrow{h}_t$ and backward LSTM output $\overleftarrow{h}_t$, which are fed to the hidden layer to yield BiLSTM's final output y_t. Its formula is as follows:

$$\begin{cases} \overrightarrow{h}_t = LSTM\left(x_t, \overrightarrow{h}_{t-1}\right) \\ \overleftarrow{h}_t = LSTM\left(x_t, \overleftarrow{h}_{t-1}\right) \\ y_t = \overrightarrow{W}\overrightarrow{h}_t + \overleftarrow{W}\overleftarrow{h}_t + b_y \end{cases} \tag{25}$$

where $\overrightarrow{W}$, $\overleftarrow{W}$, and b_y denote the forward LSTM output weight, backward LSTM output weight, and BiLSTM output bias term, respectively.

3.6. Attention mechanism

For traditional time-series data processing, capturing long-sequence relationships has long challenged RNN and LSTM. Though BiLSTM mitigate this issue, they assign equal importance to outputs across all time

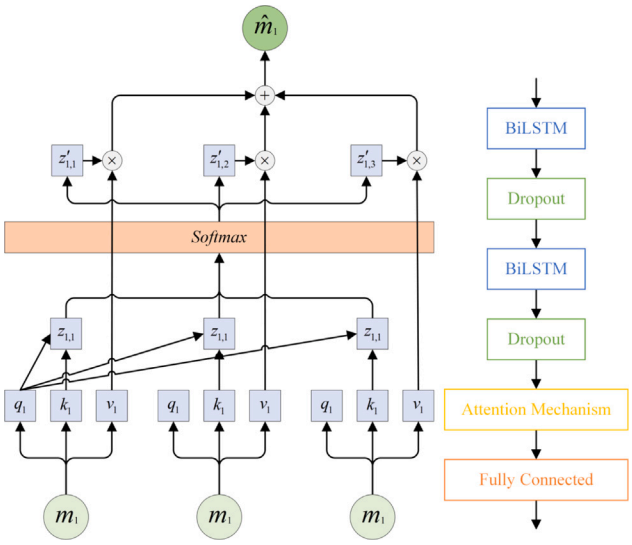


Fig. 6. Network structure of Attention mechanism and BiLSTM-Attention Mechanism.

steps—yet in practice, data at specific positions may be more critical to predictions. The attention mechanism reallocates data prediction weights, allowing results to prioritize key information [25,26].

For 1D sequence data $M = \{m_1, m_2, \ldots, m_i\}$, different weights are applied to generate distinct vectors, with its formula given as follows:

$$\begin{cases} q_i = W_q m_i \\ k_i = W_k m_i \\ v_i = W_v m_i \end{cases} \tag{26}$$

where q_i, k_i, v_i denote the query, key, and value vectors respectively; W_q, W_k, W_v are their corresponding weight matrices.

For each sequence element m_i, the attention score $z_{i,j}$ between m_i and other elements is computed to quantify its attention weight within the overall sequence. The corresponding formula is as follows:

$$z_{i,j} = q_i \cdot k_j \tag{27}$$

Normalizing the attention scores yields the attention distribution $z'_{i,j}$, with its formula given as follows:

$$z'_{i,j} = \frac{e^{z_{i,j}}}{\sum_{j=1}^{n} e^{z_{i,j}}} \tag{28}$$

The self-attention mechanism's output $\hat{M} = \{\hat{m}_1, \hat{m}_2, \ldots, \hat{m}_i\}$ is derived from a weighted sum of value vectors v_j based on $z'_{i,j}$, with its formula given as follows:

$$\hat{m}_i = \sum_{j=1}^{n} z'_{i,j} v_j \tag{29}$$

The attention mechanism's network architecture and the final BiLSTM framework is displayed in Fig. 6.

4. Improved artificial lemming algorithm

ALA is a novel bio-inspired optimization algorithm [27]. Inspired by lemming behaviors, it boasts strong exploration and exploitation capabilities, and uses an energy reduction mechanism to dynamically balance these two abilities, thus boosting its global optimization performance.

4.1. Circle chaotic mapping

To boost initial population diversity in ALA, the Circle chaotic map is used to traverse the initial population [28]. This approach benefits from the map's simple calculations, uniformly distributed iterative sequences over [0,1), and non-convergence to fixed points, effectively preventing the sequence from falling into local regions. The corresponding formula is as follows:

$$C_{i+1} = mod\left(C_i + 0.2 - \left(\frac{0.5}{2\pi}\right)\sin\left(2\pi C_i\right), 1\right) \tag{30}$$

where $mod(\cdot)$ is modulo function, C_i is ith mapping chaotic value.

Map chaotic values to the ALA initial population range, with its formula given as follows:

$$x_{i,j} = lb + C_{i,j} \cdot (ub - lb) \tag{31}$$

where $x_{i,j}$ is ALA's ith initial population individual in the jth dimension, $C_{i,j}$ is its jth dimensional ith mapping value, lb, ub is lower and upper bounds of the initial population range, respectively.

4.2. Artificial lemming algorithm

Assume that the initial population matrix of ALA is as follows:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,j} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,j} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i,1} & x_{i,2} & \cdots & x_{i,j} \end{bmatrix} \tag{32}$$

(1) Long-distance migration

To simulate population diffusion, Brownian motion and random vectors are used to boost global exploration capability, with its formula given as follows:

$$\begin{aligned} X_i(t+1) = & X_{best}(t) + F \times BM \times \left(R \times \left(X_{best}(t) - X_i(t)\right)\right) \\ & + (1 - R) \times \left(X_i(t) - X_a(t)\right) \end{aligned} \tag{33}$$

where $X_{best}(t)$ is global optimal position at time t, $X_i(t)$ is ith individual's position at time t, $X_a(t)$ is random individual's position at time t, BM is normally distributed Brownian motion, $R \in [0, 1]$ is random number, $F \in \{-1, 1\}$.

(2) Burrowing

Population positions are updated based on the current optimal solution and random individual positions, with the corresponding formula given below:

$$X_i(t+1) = X_i(t) + F \times L \times \left(X_{best}(t) - X_b(t)\right) \tag{34}$$

where $X_b(t)$ is position of a random individual distinct from $X_a(t)$, L is periodic disturbance factor.

(3) Foraging

To simulate the spiral foraging behavior of artificial lemmings, the search radius is dynamically adjusted based on distance. Its formula is as follows:

$$\begin{cases} X_i(t+1) = X_{best}(t) + F \times \varphi \times rand \times X_i(t) \\ \varphi = \chi \times (\sin(2 \times \pi \times rand) + \cos(2 \times \pi \times rand)) \\ \chi = \sqrt{\sum \left(x_{best,j}(t) - x_{i,j}(t)\right)^2} \end{cases} \tag{35}$$

where φ is spiral parameter, χ is search radius, $rand$ is random number, $x_{best,j}(t)$ is optimal individual's jth dimensional position, $x_{i,j}(t)$ is ith individual's jth dimensional position.

(4) Evading predators

When artificial lemmings encounter danger, they flee back to their burrows immediately and adopt deceptive behaviors to evade pursuit, with the corresponding formula given below.

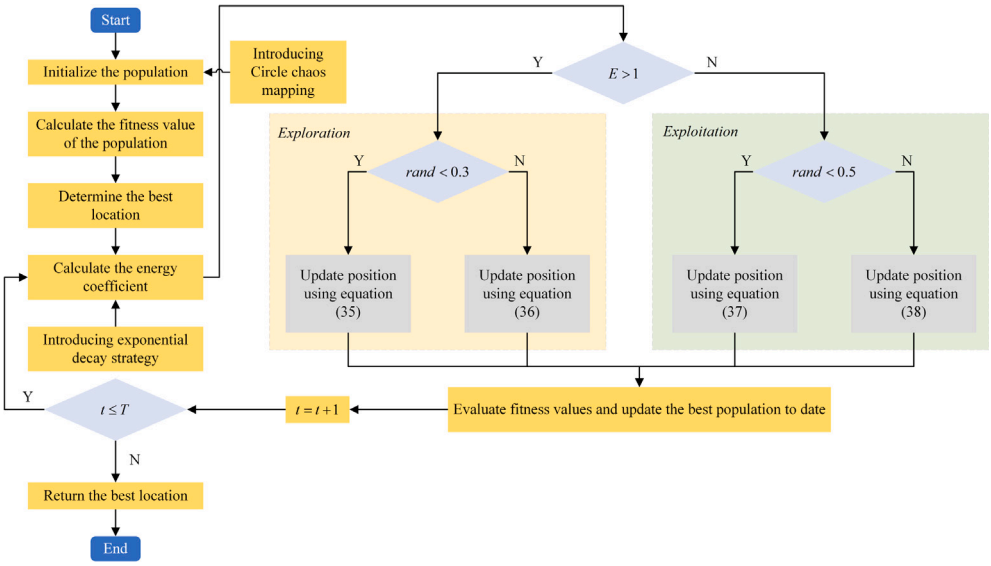


Fig. 7. Flowchart of IALA.

$$X_i(t+1) = X_{best}(t) + F \times 2 \times \left(1 - \frac{t}{T}\right) \times levy \times \left(X_{best}(t) - X_i(t)\right) \tag{36}$$

where T is maximum iteration count, $levy$ is Levy flight strategy.

(5) Energy control mechanism

To balance exploration and exploitation, ALA incorporates an energy factor: artificial lemmings migrate or burrow when energy is sufficient, and forage or evade predators otherwise. The energy factor's calculation formula is given below.

$$E(t) = 4 \times \arctan\left(1 - \frac{t}{T}\right) \times \ln\left(\frac{1}{rand}\right) \tag{37}$$

When $E(t) > 1$, migration or burrowing behaviors are executed; otherwise, foraging or evasion behaviors are implemented.

To boost early-stage global exploration, enhance late-stage local exploitation, ensure energy factor stability and continuity during decay, and improve algorithm convergence accuracy, an exponential decay strategy is introduced for the energy factor, with the corresponding formula given below.

$$E(t) = e^{-\alpha \times \left(\frac{t}{T}\right)^{\beta}} \times \ln\left(\frac{1}{rand}\right) \tag{38}$$

where α, β are adjustable parameters for controlling decay rate and curve shape.

The flowchart of IALA is shown in Fig. 7.

5. Model of TCN-BiLSTM

This paper adopts a parallel prediction method combining the TCN and BiLSTM models. By harnessing TCN's strengths (strong parallel computing capability and superior long-distance sequence relationship capture) and BiLSTM's advantages (bidirectional information fusion and suitability for locally dynamically complex data processing), it achieves the "global dependency capture + local time-series modeling" synergy. The two models' predictions are compared with the original signal; the result with smaller error at each sampling point is selected and stacked to generate the final prediction output. The per-point error index formula is given below.

$$\Delta(t) = \left|y_t - \hat{y}_t\right| \tag{39}$$

where y_t is original signal sampling point, $\hat{y}_t$ is TCN/BiLSTM-predicted data point.

Finally, based on the point-wise error comparison selective ensemble strategy, the stacking formula for TCN-BiLSTM model prediction points is given below.

$$y_{final}(t) = \begin{cases} y_{BiLSTM}(t) & \Delta_{BiLSTM}(t) < \Delta_{TCN}(t) \\ y_{TCN}(t) & \Delta_{BiLSTM}(t) \geq \Delta_{TCN}(t) \end{cases} \tag{40}$$

where $y_{final}(t)$ is TCN-BiLSTM model's tth prediction point, $y_{BiLSTM}(t)$ is BiLSTM model's tth prediction point, $y_{TCN}(t)$ is TCN model's tth prediction point.

The TCN-BiLSTM model's prediction process is illustrated in Fig. 8, with its specific steps outlined below.

Signal preprocessing:

Step 1: Collect the original voltage data of PEMFC;
Step 2: Original voltage data are decomposed into multiple IMFs via VMD, with decomposition into independent IMFs determined by mutual information and Pearson correlation coefficient;
Step 3: Each IMF is denoised via KF to eliminate high-oscillation-amplitude noise;
Step 4: KF-denoised signals are further denoised via MF to eliminate small sharp-point noises and yield the final reconstructed signal.

TCN-BiLSTM prediction:

Step 1: Circle chaotic map is integrated into ALA to optimize the initial population, and an exponential decay strategy is added to the energy factor, forming IALA;
Step 2: IALA optimizes BiLSTM weights to identify optimal weight values, then integrates with the attention mechanism for prediction;
Step 3: Using TCN for prediction;
Step 4: BiLSTM and TCN predictions are compared by error; data with smaller errors are selected and stacked to yield the final prediction result.

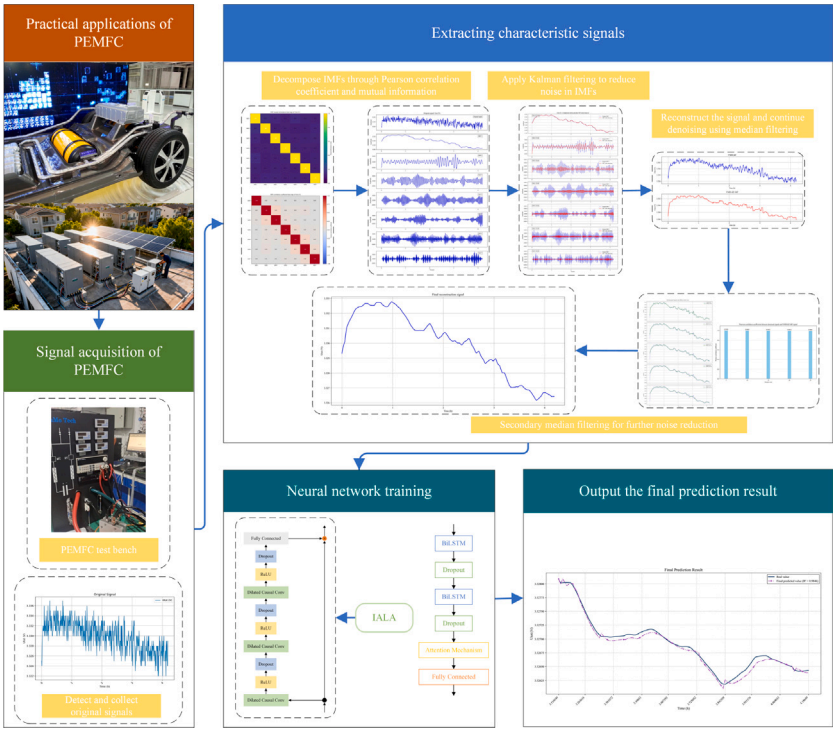


Fig. 8. Model prediction process of TCN-BiLSTM.

Table 2
Comparison of prediction results of various models.

Prediction model	MSE	MAE	R^2
TCN	2×10^{-8}	1.0971×10^{-4}	90.62%
BiLSTM	4×10^{-8}	1.3574×10^{-4}	81.80%
PSO-BiLSTM	2×10^{-8}	1.4167×10^{-4}	89.04%
GWO-BiLSTM	2×10^{-8}	1.0197×10^{-4}	90.61%
ALA-BiLSTM	2×10^{-8}	1.0452×10^{-4}	92.37%
IALA-BiLSTM	9.2463×10^{-9}	7.6515×10^{-5}	95.79%
TCN (With residual connection)	2.7948×10^{-9}	4.0271×10^{-5}	98.73%
IALA-BiLSTM-Attention	7.1253×10^{-9}	6.6303×10^{-5}	96.76%
TCN-BiLSTM	1.4211×10^{-9}	2.6636×10^{-5}	99.35%

6. Experimental analysis

To verify the model's superior performance, this experiment adopts MSE, mean absolute error (MAE), and coefficient of determination (R^2) as comprehensive evaluation metrics. Among them, (R^2) acts as a prediction accuracy indicator for intuitive accuracy reflection, with its formula given below.

$$\begin{cases} \sigma_{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2 \\ \sigma_{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_i - y_i\right| \\ \sigma_{R^2} = 1 - \frac{\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2}{\sum_{i=1}^{n}\left(\bar{y} - y_i\right)^2} \end{cases} \tag{41}$$

where $\bar{y}$ is original signal average value.

The comparison of prediction results is shown in Table 2 below.

Existing models are analyzed in Fig. 9, while the improved model is analyzed in Figs. 10 and 11.

Adding residual connections to the TCN boosted its accuracy from 90.62% to 98.73% with nearly no delay in iterative convergence. This improvement relies on residual connections' gradient stabilization and feature reuse, which resolve TCN's training bottlenecks and better adapt it to the long-sequence, high-interference nature of PEMFC voltage signals. Analysis of prediction error distribution, histograms, and scatter plots shows errors range within (−0.00015 to 0.00015), follow a Gaussian distribution, and predicted points closely match ideal values—proving the error is negligible relative to real signals and predictions are highly accurate.

BiLSTM's loss curve converged as early as the 5th iteration, likely due to premature local convergence during training.

PSO-BiLSTM converges rapidly but exhibits post-convergence fluctuations, leading to poor convergence stability. GWO-BiLSTM is both slower to converge and less stable than PSO-BiLSTM. These drawbacks impede the minimization of loss function error and cannot satisfy high-precision prediction requirements.

IALA outperforms ALA significantly in both convergence speed and stability, verifying the effectiveness of ALA's improvement measures.

Multiple experiments show that IALA stably converges within 20 iterations, with minimal prediction errors (−0.00017 to 0.0001) following a Gaussian distribution. Its predicted points closely match ideal

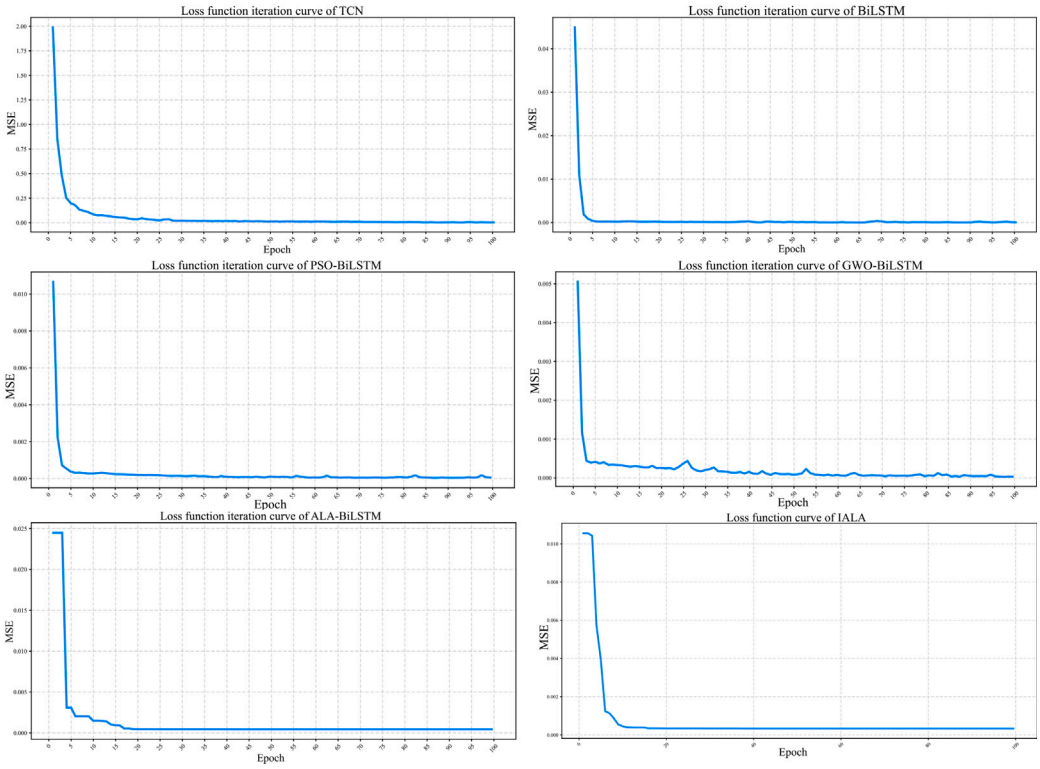


Fig. 9. Analysis of existing model loss functions.

values, and prediction accuracy rises from 81.80% to 96.76%, marking a notable boost in predictive performance.

The prediction results of each model are presented in Fig. 12.

The final TCN-BiLSTM prediction result stems from error-based selection of improved TCN and BiLSTM outputs. Fusing the two models' predictions enables the final result to inherit their respective advantages; prediction accuracy rises further from the individual models' 98.73% and 96.76% to 99.35%, boosting accuracy and verifying the proposed novel model's feasibility and superiority.

Remark 3. The strengths of the proposed hybrid architecture can be summarized as follows: TCN can effectively capture cross-time-scale feature correlations in sequences by adjusting the receptive field without raising model complexity, compensating for BiLSTM's computational efficiency limitations [29], while BiLSTM's bidirectional recurrent structure enables learning sequence dependencies from both past and future perspectives and excels at capturing short-sequence feature relationships. The two components thus form a complementary framework: TCN specializes in representing abrupt local variations such as instantaneous load fluctuations, whereas BiLSTM prioritizes global contextual information to effectively correlate performance evolution around such fluctuations, and integrating these complementary advantages mitigates the inherent biases of standalone models when extracting either local or global features.

Remark 4. In addition, adding residual connections and dropout regularization to the model effectively mitigates overfitting and underfitting, enhancing model robustness. For complex signal processing, this paper extracts pure feature signals stepwise via multiple filtering algorithms; by leveraging the strengths of various filters, it eliminates noise without compromising the original feature correlations.

7. Conclusion

To tackle three key challenges in PEMFC RUL prediction—low accuracy, standalone deep learning model limitations, and noise interference in feature extraction—this paper proposes a parallel TCN-BiLSTM model. The model reaches 99.35% prediction accuracy, outperforming comparative models while boosting predictive capability without extending training time. It should be noted that the model is currently verified only based on the FCLAB dataset and has not been tested under actual complex operating conditions such as extreme temperature/humidity and sudden load changes, which constitutes a limitation to its direct industrial application. It lays a reliable foundation for PEMFC predictive maintenance, driving the field to shift from single-model approaches to multi-model complementary modeling. Its AI-driven computational framework can also be extended to RUL and performance prediction for other advanced energy devices (e.g., perovskite solar cells, solid-state lithium batteries), supporting AI-enabled integrated materials and device design. Based on this research, future work will focus on multi-physical quantity fusion models, vehicle-mounted lightweight deployment, and cross-condition domain-adaptive transfer learning to enhance generalization and advance PEMFC industrialization.

CRediT authorship contribution statement

Jian Jiang: Writing – original draft, Methodology, Data curation. **Dongsheng Du:** Supervision, Methodology. **Lin Su:** Validation, Software. **Yifan Du:** Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that there are no conflicts of interest in relation to the manuscript. The authors confirm that the results and interpretations reported in the manuscript are original and have not been plagiarized.

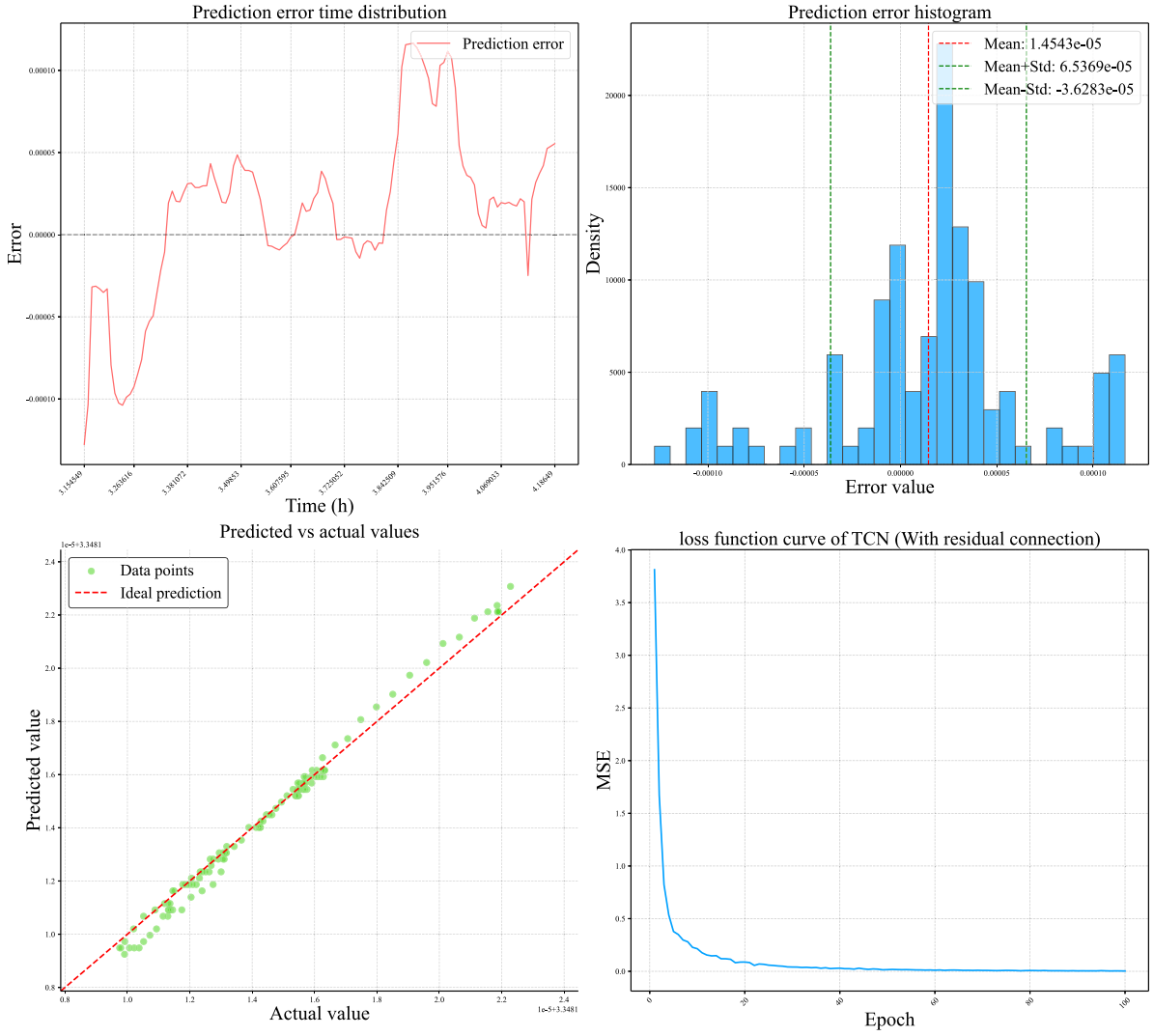


Fig. 10. Analysis chart of TCN (with residual connection) indicators.

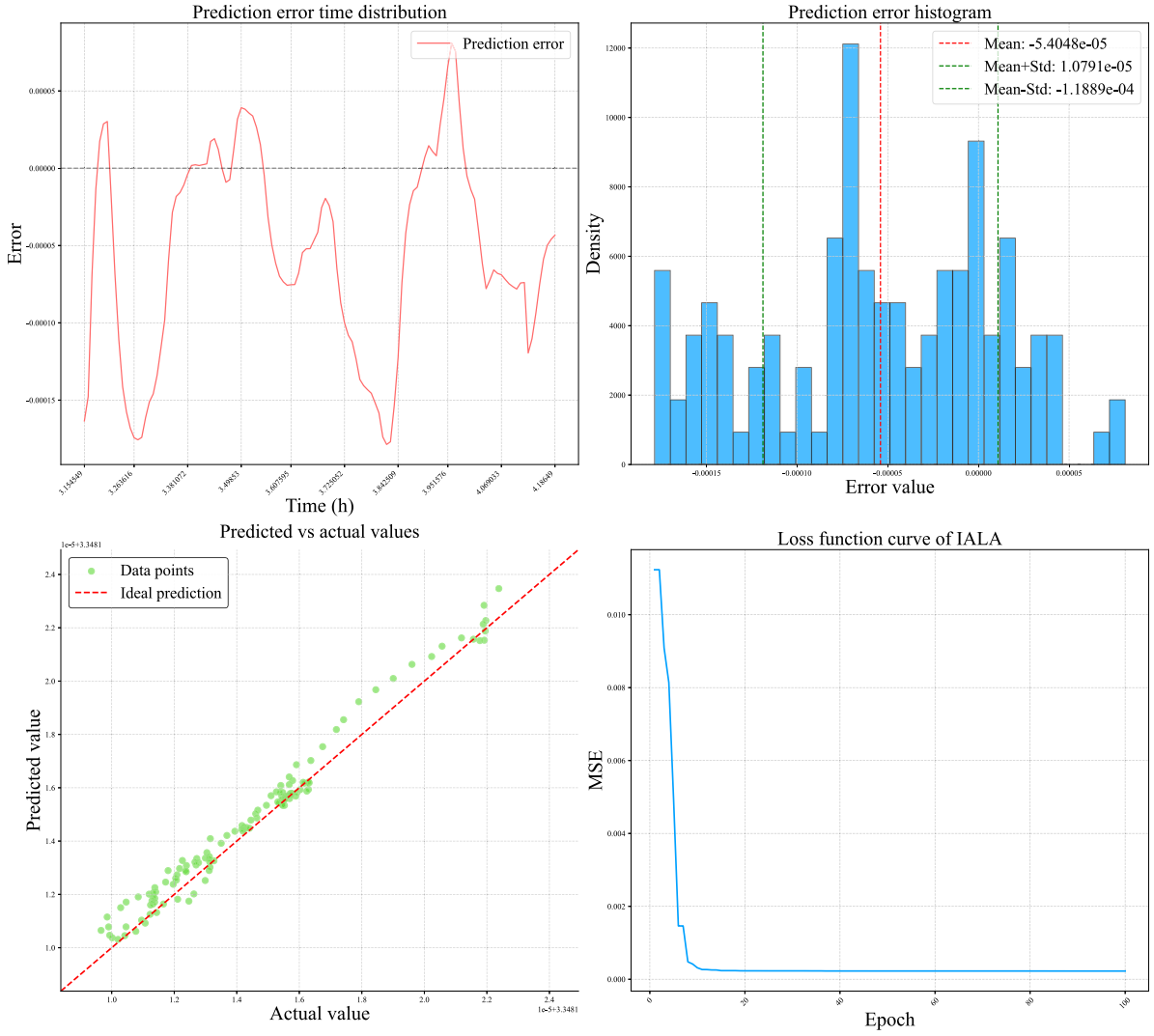


Fig. 11. Analysis chart of IALA-BiLSTM-Attention indicators.

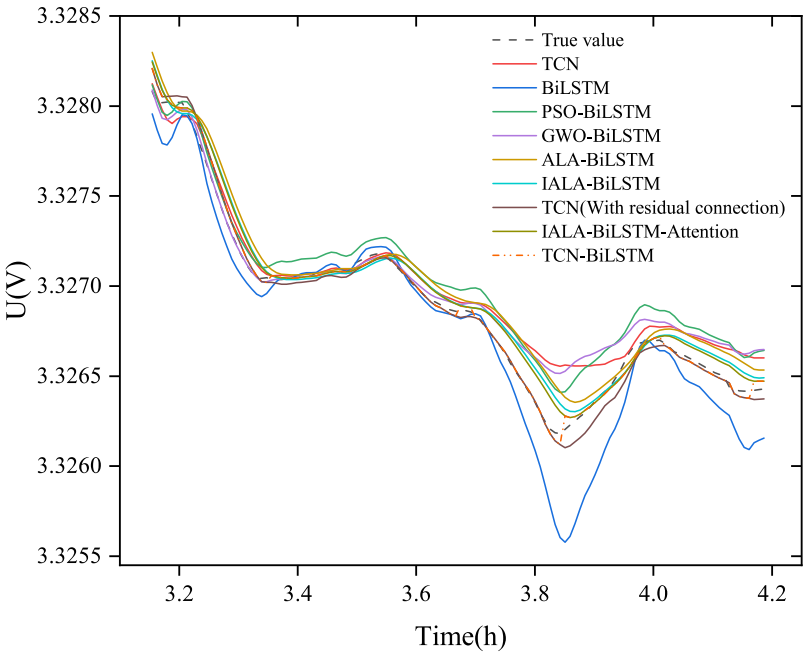


Fig. 12. Prediction results of each model.

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